AIR BNB DATABASE

Nontobeko Ncube

1 User

Purpose: Guest and host management
Business Question: What is our user demographic breakdown and account health status?

Test Query:

```
SELECT AccountStatus,
COUNT(*) as UserCount,
    AVG(YEAR(CURDATE()) -
YEAR(DateOfBirth)) as AvgAge,
    MIN(CreationDate) as
FirstUser,
    MAX(CreationDate) as
LatestUser
FROM User
GROUP BY AccountStatus;
```

```
SELECT AccountStatus, COUNT(*) as UserCount,
       AVG(YEAR(CURDATE()) - YEAR(DateOfBirth)) as AvgAge,
       MIN(CreationDate) as FirstUser,
       MAX(CreationDate) as LatestUser
FROM testdb.user
GROUP BY AccountStatus;
```

user 1

SELECT AccountStatus, COUNT(*) as UserCount, AVG(YE

Enter a SQL expression to filter results (use Ctrl+Space)

	AccountStatus	UserCount	AvgAge	FirstUser	LatestUser
1	ACTIVE	25	35.72	2024-01-10 09:30:00.0	2024-03-12 12:10:00.0

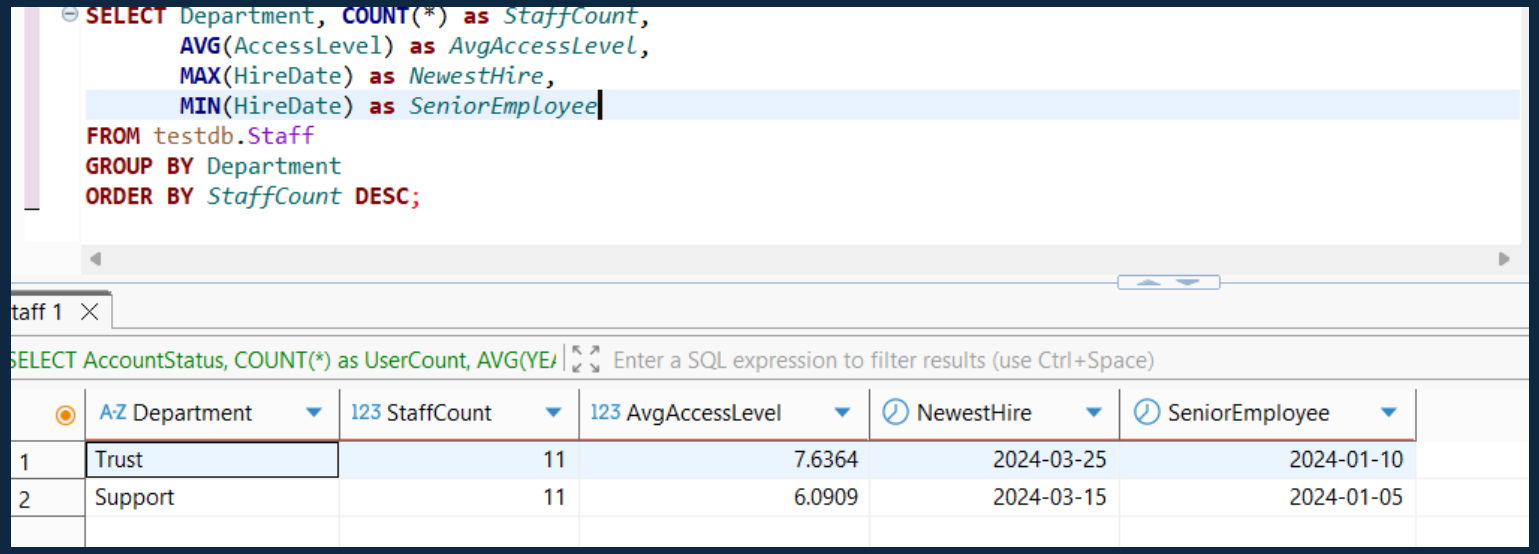
2 Staff Table

Purpose: Stores company staff information for verification processes and platform administration

Business Question: How is our support team structured and what are the experience levels across departments?

Query:

```
SELECT Department, COUNT(*) as StaffCount,
    AVG(AccessLevel) as AvgAccessLevel,
    MAX(HireDate) as NewestHire,
    MIN(HireDate) as SeniorEmployee
FROM Staff
GROUP BY Department
ORDER BY StaffCount DESC;
```



	Department	StaffCount	AvgAccessLevel	NewestHire	SeniorEmployee
1	Trust	11	7.6364	2024-03-25	2024-01-10
2	Support	11	6.0909	2024-03-15	2024-01-05

3 Languages Table

Purpose: Master lookup table for supported languages on the platform

Business Question: Which languages do our hosts speak and how does this support international guests?

Query:

```
SELECT LanguageName, LanguageCode,
    COUNT(hl.HostID) as HostSpeakers
FROM Languages l
LEFT JOIN Host_Languages hl ON l.LanguageID = hl.LanguageID
GROUP BY l.LanguageID
ORDER BY HostSpeakers DESC;
```

```
SELECT LanguageName, LanguageCode,
       COUNT(hl.HostID) as HostSpeakers
FROM Languages l
LEFT JOIN Host_Languages hl ON l.LanguageID = hl.LanguageID
GROUP BY l.LanguageID
ORDER BY HostSpeakers DESC;
```

guages 1

LECT Department, COUNT(*) as StaffCount, AVG(Acce | Enter a SQL expression to filter resu

LanguageName	LanguageCode	HostSpeakers
Zephyrian Common	ZE	10
Skyborn	SK	3
Shadowkin	SH	2
Earthen	EA	2
Windish	WI	2
Crystal Tongue	CR	1
Dwarvish	DW	1
Draconic	DR	1
Aquatic	AQ	1
Forestspeak	FR	1
Firetongue	FI	1
Voidian	VO	1
Stellar	ST	1
Moonkin	MO	1
Elvish	EL	0
Lightspeak	LI	0
Solarian	SU	0
Temporal	TI	0
Spiritish	SP	0
Arcane	MA	0
Runic	RU	0
Mystic	MY	0

4 ListingAddress Table

Purpose: Physical locations of listings with GPS coordinates for mapping and search

Business Question: What is our geographic coverage and where are listings concentrated?

Query:

```
SELECT Country, State, COUNT(*) as ListingCount,
      AVG(Latitude) as AvgLat, AVG(Longitude) as AvgLong
FROM ListingAddresses
GROUP BY Country, State
ORDER BY ListingCount DESC;
```

```
SELECT Country, State, COUNT(*) as ListingCount,
       AVG(Latitude) as AvgLat, AVG(Longitude) as AvgLong
FROM ListingAddresses
GROUP BY Country, State
ORDER BY ListingCount DESC;
```

listingaddresses 1

SELECT LanguageName, LanguageCode, COUNT(hl.Host | Enter a SQL expression to filter results (use Ctrl+Space)

	Country	State	ListingCount	AvgLat	AvgLong
1	Upper Zephyria	Sky Kingdom	5	47.7900984	-69.0790094
2	Western Zephyria	Forest Realm	4	43.64478325	-71.53392225
3	Eastern Zephyria	Fire Realm	3	41.3827116667	-75.271604
4	Southern Zephyria	Water Kingdom	3	39.56786	-77.4567856667
5	Northern Zephyria	Crystal Kingdom	2	45.4011785	-73.4012285
6	Radiant Zephyria	Light Kingdom	2	46.8956785	-70.8395675
7	Hidden Zephyria	Void Realm	2	40.5067895	-76.4006785
8	Shadow Zephyria	Night Realm	1	44.789012	-72.678901
9	Central Zephyria	Mountain Empire	1	42.901234	-74.890123
10	Eternal Zephyria	Time Realm	1	45.789012	-73.678901
11	Resonant Zephyria	Music Realm	1	44.890123	-72.789012

5 PropertyType Table

Purpose: Categories for different types of accommodation properties

Business Question: Which property types are most popular on our platform?

Query:

```
SELECT pt.TypeName,
pt.TypeDescription,
    COUNT(p.PropertyID) as
PropertyCount
FROM PropertyType pt
LEFT JOIN Property p ON
pt.PropertyTypeID =
p.PropertyTypeID
GROUP BY pt.PropertyTypeID
ORDER BY PropertyCount
DESC;
```

```
SELECT pt.TypeName, pt.TypeDescription,
       COUNT(p.PropertyID) as PropertyCount
FROM propertytype pt
LEFT JOIN Property p ON pt.PropertyTypeID = p.PropertyTypeID
GROUP BY pt.PropertyTypeID
ORDER BY PropertyCount DESC;
```

propertytype 1

SELECT pt.TypeName, pt.TypeDescription, COUNT(p.Pro | Enter a SQL expression to filter results (use Ctrl+Space)

	TypeName	TypeDescription	PropertyCount
1	Underwater Dome	Magical air-filled domes beneath the waves	2
2	Mountain Fortress	Stone fortresses carved into mountain peaks	2
3	Phoenix Nest	High-altitude nests adapted for travelers	2
4	Crystal Palace	Majestic ice and crystal structures in the Crystal Kingdom	1
5	Tree Mansion	Luxury treehouses in the ancient forests	1
6	Cloud Castle	Floating castles in the sky realm	1
7	Dragon Cave	Converted dragon lairs with modern amenities	1
8	Wizard Tower	Multi-story towers with magical conveniences	1
9	Royal Villa	Aristocratic villas in kingdom capitals	1
10	Mystical Yurt	Nomadic dwellings with portal connections	1
11	Mushroom House	Giant mushroom dwellings in magical forests	1
12	Ice Igloo	Magical ice structures that never melt	1
13	Earth Lodge	Underground lodges with crystal lighting	1
14	Star Observatory	Celestial viewing platforms with lodging	1
15	Elemental Sanctuary	Dwellings attuned to specific magical elements	1
16	Portal Inn	Multi-dimensional inns with realm access	1
17	Healing Springs	Restorative hot spring accommodations	1
18	Time Capsule	Temporal dwellings with unique time effects	1
19	Desert Oasis	Enchanted oasis compounds in magical deserts	0
20	Fairy Cottage	Miniature cottages enhanced with size magic	0
21	Enchanted Cabin	Cozy cabins with protective enchantments	0

6 ExperienceTypes

Purpose: Categories for different types of experiences offered on the platform

Business Question: What types of experiences do we offer and which categories are most developed?

Query:

```
SELECT et.TypeName,
et.TypeDescription,
    COUNT(e.ExperienceID) as
ExperienceCount
FROM ExperienceTypes et
LEFT JOIN Experience e ON
et.ExperienceTypeID =
e.ExperienceTypeID
GROUP BY et.ExperienceTypeID
ORDER BY ExperienceCount
DESC;
```

```
SELECT et.TypeName, et.TypeDescription,
       COUNT(e.ExperienceID) as ExperienceCount
FROM ExperienceTypes et
LEFT JOIN Experience e ON et.ExperienceTypeID = e.ExperienceTypeID
GROUP BY et.ExperienceTypeID
ORDER BY ExperienceCount DESC;
```

experiencetypes 1

SELECT pt.TypeName, pt.TypeDescription, COUNT(p.Pro

Enter a SQL expression to filter results (use Ctrl+Space)

	TypeName	TypeDescription	ExperienceCount
1	Elemental Mastery	Basic control of fire, water, earth, or air	2
2	Spirit Channeling	Communicate with spirits of departed heroes	2
3	Healing Arts	Learn magical healing techniques	2
4	Void Meditation	Safe exploration of void realm consciousness	2
5	Dragon Riding	Guided flights on friendly dragons across kingdoms	1
6	Spell Weaving	Learn basic magic from certified wizards	1
7	Crystal Mining	Supervised mining expeditions in crystal caves	1
8	Forest Communion	Connect with ancient tree spirits and forest magic	1
9	Underwater Exploration	Magical breathing apparatus for ocean depths	1
10	Sky Walking	Levitation lessons and cloud kingdom tours	1
11	Potion Brewing	Hands-on alchemy with master alchemists	1
12	Phoenix Watching	Witness the legendary phoenix rebirth cycle	1
13	Time Travel	Supervised journeys to historical moments	1
14	Unicorn Encounter	Meet and feed magical unicorns in sacred groves	1
15	Portal Crafting	Learn to create temporary travel portals	1
16	Starlight Dancing	Traditional celestial dances under magical stars	1
17	Rune Reading	Ancient divination with mystical rune stones	1
18	Shapeshifting	Temporary animal transformation experiences	1
19	Mind Reading	Telepathy training with ethical guidelines	1
20	Fairy Diplomacy	Negotiation and communication with fairy courts	1

7 ServiceTypes

Purpose: Categories for different types of services provided by hosts

Business Question: What service categories are available and where should we focus growth?

Query:

```
SELECT st.TypeName,
st.TypeDescription,
    COUNT(s.ServiceID) as
ServiceCount
FROM ServiceTypes st
LEFT JOIN Service s ON
st.ServiceTypeID =
s.ServiceTypeID
GROUP BY st.ServiceTypeID
ORDER BY ServiceCount DESC;
```

```
SELECT st.TypeName, st.TypeDescription,
       COUNT(s.ServiceID) as ServiceCount
FROM ServiceTypes st
LEFT JOIN Service s ON st.ServiceTypeID = s.ServiceTypeID
GROUP BY st.ServiceTypeID
ORDER BY ServiceCount DESC;
```

servicetypes 1

SELECT et.TypeName, et.TypeDescription, COUNT(e.Exp

Enter a SQL expression to filter results (use Ctrl+Spa

	TypeName	TypeDescription	ServiceCount
1	Healing Magic	Magical healing services for injuries and ailments	2
2	Spell Casting	Custom magic spells performed by certified mages	1
3	Portal Transportation	Instant travel between designated portal points	1
4	Enchanted Photography	Magical portraits that move and speak	1
5	Weather Modification	Localized weather control for events	1
6	Protective Wards	Magical security systems for homes and events	1
7	Dream Crafting	Custom dream experiences while sleeping	1
8	Memory Enhancement	Magical improvement of memory and learning	1
9	Language Telepathy	Instant translation through magical telepathy	1
10	Familiar Summoning	Temporary magical creature companions	1
11	Blessing Ceremonies	Magical blessings for special occasions	1
12	Curse Removal	Professional removal of harmful magical curses	1
13	Astral Projection	Guided out-of-body spiritual experiences	1
14	Elemental Cleaning	Magical cleaning using elemental forces	1
15	Illusion Entertainment	Magical illusions for parties and events	1
16	Prophecy Reading	Professional fortune telling and future sight	1
17	Magical Cooking	Enchanted meals with beneficial magical effects	1
18	Plant Growth	Accelerated growth magic for gardens	1
19	Crystal Charging	Magical energy infusion for crystal devices	1
20	Soul Bonding	Magical ceremonies for deep relationships	1
21	Dimensional Storage	Expanded space magic for personal belongings	1

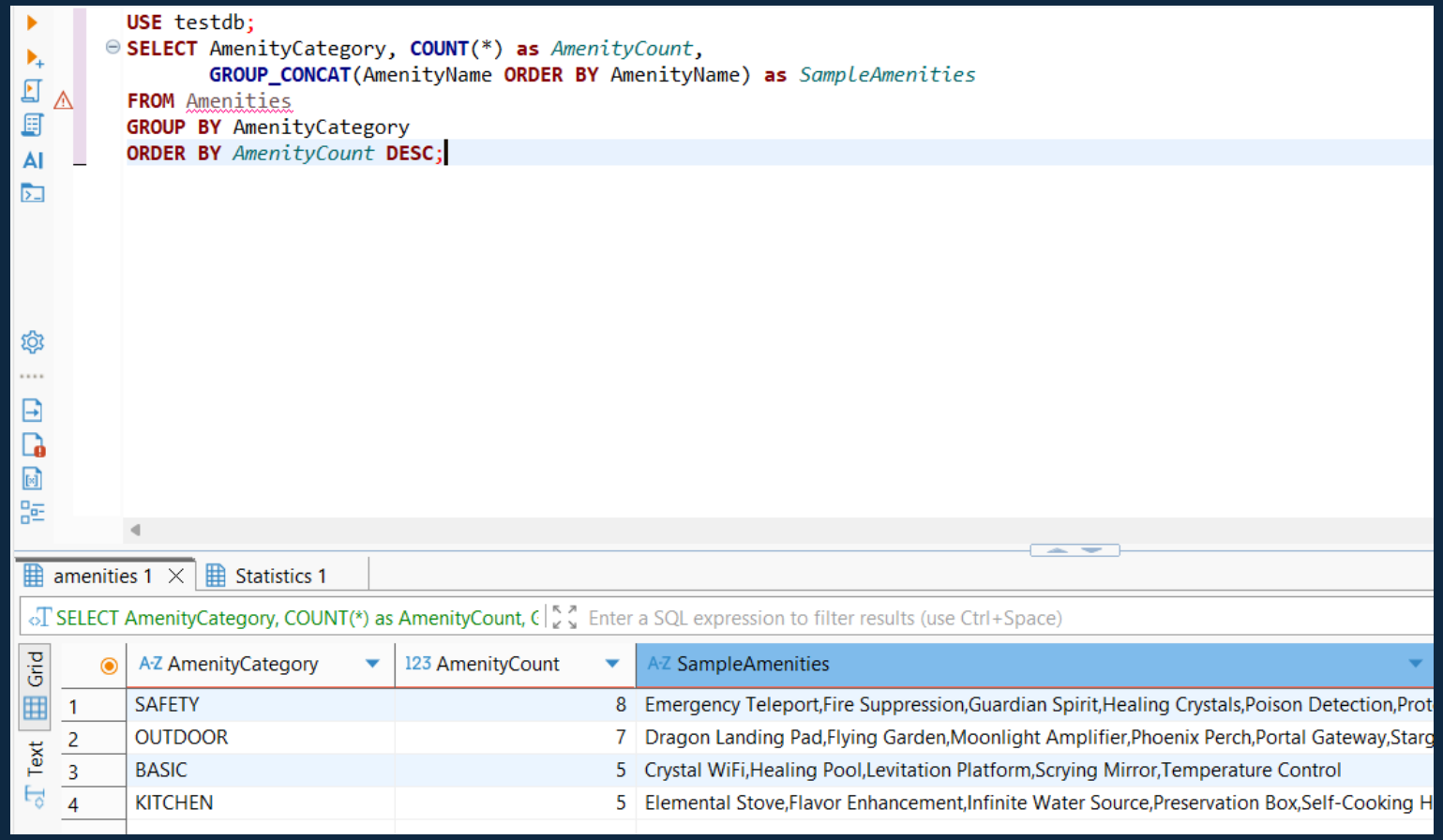
8 Amenities

Purpose: Master lookup for amenities and features available across listings

Business Question: How comprehensive is our amenity offering across different categories?

Query:

```
SELECT AmenityCategory, COUNT(*) as AmenityCount,
    GROUP_CONCAT(AmenityName ORDER BY AmenityName LIMIT 5) as SampleAmenities
FROM Amenities
GROUP BY AmenityCategory
ORDER BY AmenityCount DESC;
```



9 SavedPaymentDetails

Purpose: User payment methods and cards for seamless booking transactions

Business Question: What payment methods do users prefer and how healthy is our payment portfolio?

Query:

```
SELECT CardBrand, COUNT(*) as UserCount,
    AVG(ExpiryYear - YEAR(CURDATE())) as AvgYearsToExpiry,
    SUM(CASE WHEN IsDefault = 1 THEN 1 ELSE 0 END) as PrimaryCards
FROM SavedPaymentDetails
WHERE IsActive = 1
GROUP BY CardBrand
ORDER BY UserCount DESC;
```

```
SELECT CardBrand, COUNT(*) as UserCount,
       AVG(ExpiryYear - YEAR(CURDATE())) as AvgYearsToExpiry,
       SUM(CASE WHEN IsDefault = 1 THEN 1 ELSE 0 END) as PrimaryCards
FROM SavedPaymentDetails
WHERE IsActive = 1
GROUP BY CardBrand
ORDER BY UserCount DESC;
```

savedpaymentdetails 1

SELECT AmenityCategory, COUNT(*) as AmenityCount, C

Enter a SQL expression to filter results (use Ctrl+Space)

	CardBrand	UserCount	AvgYearsToExpiry	PrimaryCards
1	Celestial Bank	3	1	3
2	Water Flow	3	-0.6667	3
3	Crystal Bank	2	-1	2
4	Void Exchange	2	0.5	2
5	Fire Mint	2	0.5	2
6	Lunar Vault	1	0	1
7	Dwarf Treasury	1	-1	1
8	Sky Treasury	1	-1	1
9	Stone Vault	1	0	1
10	Forest Credit	1	0	1
11	Harmony Bank	1	-1	1
12	Shadow Mint	1	0	1
13	Mystic Vault	1	1	1

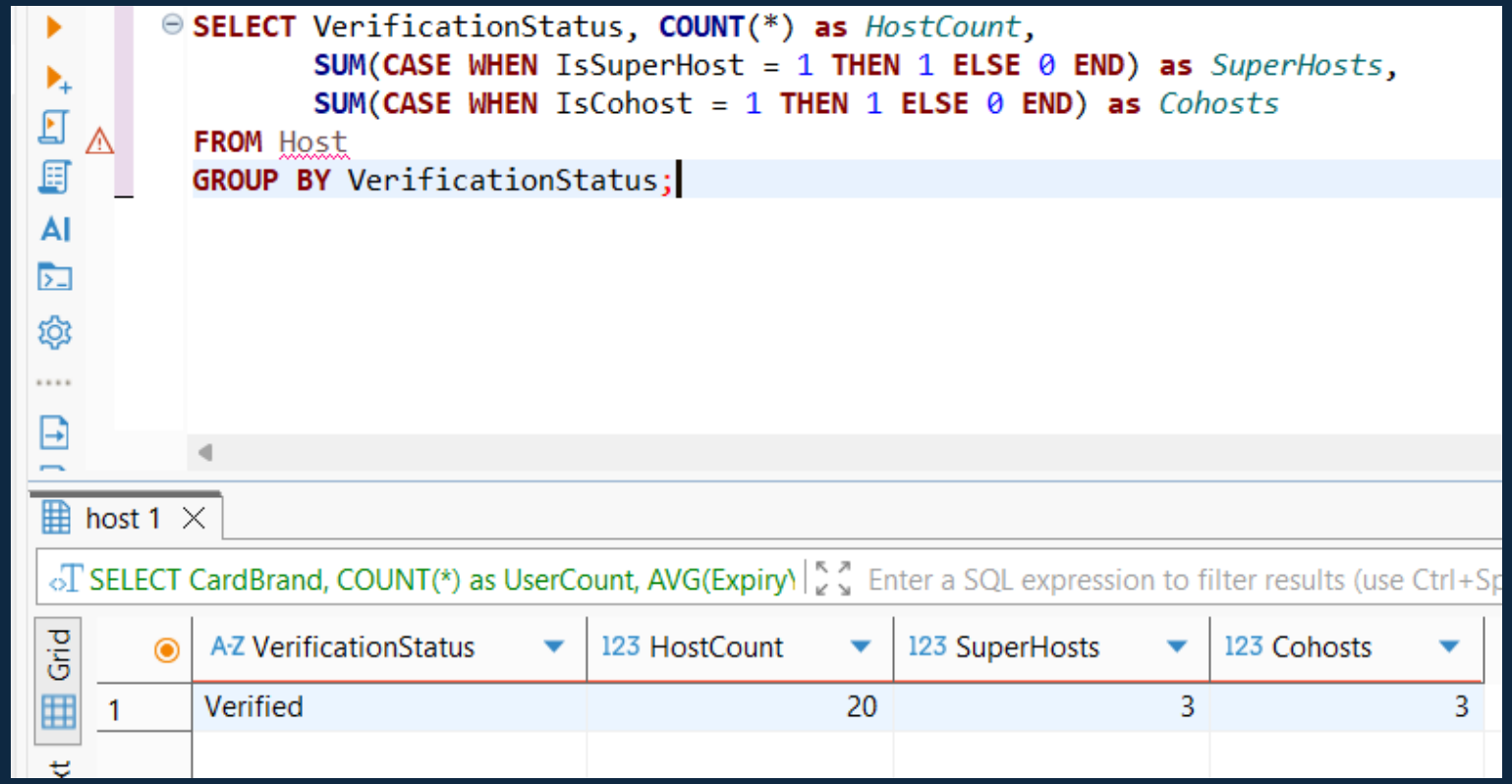
10 Host

Purpose: Host specialization with verification status and performance metrics

Business Question: What is our host verification success rate and quality metrics?

Query:

```
SELECT VerificationStatus,
COUNT(*) as HostCount,
    SUM(CASE WHEN IsSuperHost
= 1 THEN 1 ELSE 0 END) as
SuperHosts,
    SUM(CASE WHEN IsCohost = 1
THEN 1 ELSE 0 END) as Cohosts
FROM Host
GROUP BY VerificationStatus;
```



11 Addresses

Purpose: User addresses for billing, shipping, and location-based services

Business Question: Where are our users located and what address types do they provide?

Query:

```
SELECT AddressType, Country, COUNT(*) as AddressCount
FROM Addresses
GROUP BY AddressType, Country
ORDER BY AddressCount DESC
LIMIT 10;
```

```
SELECT AddressType, Country, COUNT(*) as AddressCount
FROM Addresses
GROUP BY AddressType, Country
ORDER BY AddressCount DESC
LIMIT 10;
```

addresses 1

SELECT VerificationStatus, COUNT(*) as HostCount, SUM | Enter a SQL expression

	AddressType	Country	AddressCount
1	Home	Upper Zephyria	4
2	Home	Southern Zephyria	4
3	Home	Hidden Zephyria	3
4	Home	Shadow Zephyria	2
5	Home	Eastern Zephyria	2
6	Home	Western Zephyria	2
7	Home	Northern Zephyria	2
8	Home	Radiant Zephyria	2
9	Home	Eternal Zephyria	2
10	Home	Central Zephyria	1

12 TaxDocuments

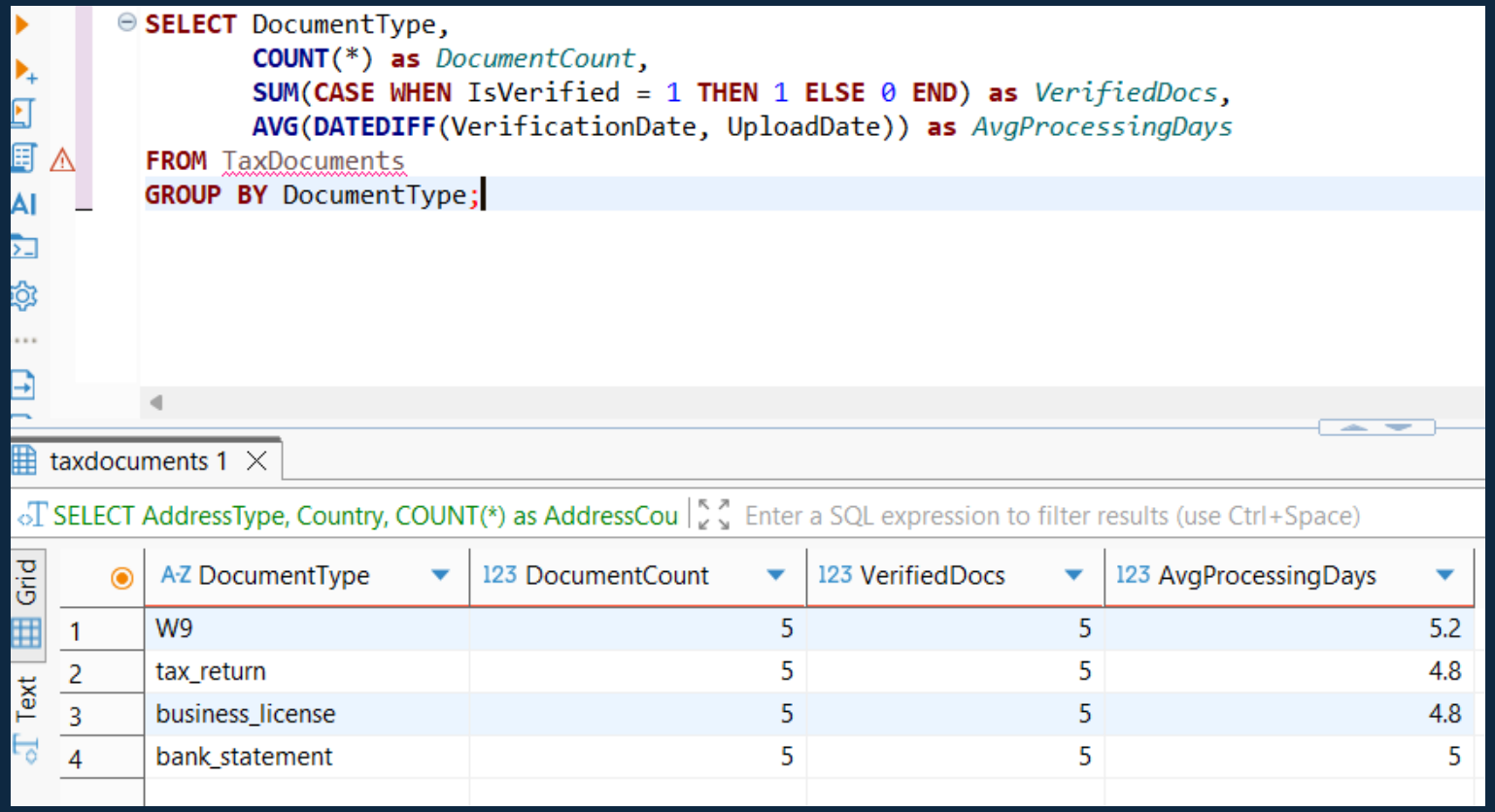
Purpose: Host tax compliance documentation for earnings reporting

Business Question: How compliant are our hosts with tax documentation and how efficient is our processing?

Query:

```
SELECT DocumentType,
    COUNT(*) as DocumentCount,
    SUM(CASE WHEN IsVerified = 1 THEN 1 ELSE 0 END) as VerifiedDocs,

AVG(DATEDIFF(VerificationDate, UploadDate)) as AvgProcessingDays
FROM TaxDocuments
GROUP BY DocumentType;
```



13 HostLanguages

Purpose: Many-to-many relationship tracking host language skills and proficiency

Business Question: What language skills do our hosts have and at what proficiency levels?

Query:

```
SELECT l.LanguageName, hl.ProficiencyLevel, COUNT(*) as HostCount
FROM Host_Languages hl
JOIN Languages l ON hl.LanguageID = l.LanguageID
GROUP BY l.LanguageID, hl.ProficiencyLevel
ORDER BY HostCount DESC;
```

```
SELECT l.LanguageName, hl.ProficiencyLevel, COUNT(*) as HostCount
FROM Host_Languages hl
JOIN Languages l ON hl.LanguageID = l.LanguageID
GROUP BY l.LanguageID, hl.ProficiencyLevel
ORDER BY HostCount DESC;
```

languages(+) 1

SELECT DocumentType, COUNT(*) as DocumentCount, S | Enter a SQL expression to filter results

	LanguageName	ProficiencyLevel	HostCount
1	Zephyrian Common	Fluent	10
2	Skyborn	Native	1
3	Windish	Fluent	1
4	Shadowkin	Native	1
5	Moonkin	Native	1
6	Dwarvish	Native	1
7	Earthen	Fluent	1
8	Stellar	Native	1
9	Skyborn	Basic	1
10	Voidian	Native	1
11	Shadowkin	Fluent	1
12	Aquatic	Native	1
13	Windish	Native	1
14	Skyborn	Fluent	1
15	Firetongue	Native	1
16	Draconic	Basic	1
17	Earthen	Native	1
18	Forestspeak	Fluent	1
19	Crystal Tongue	Native	1

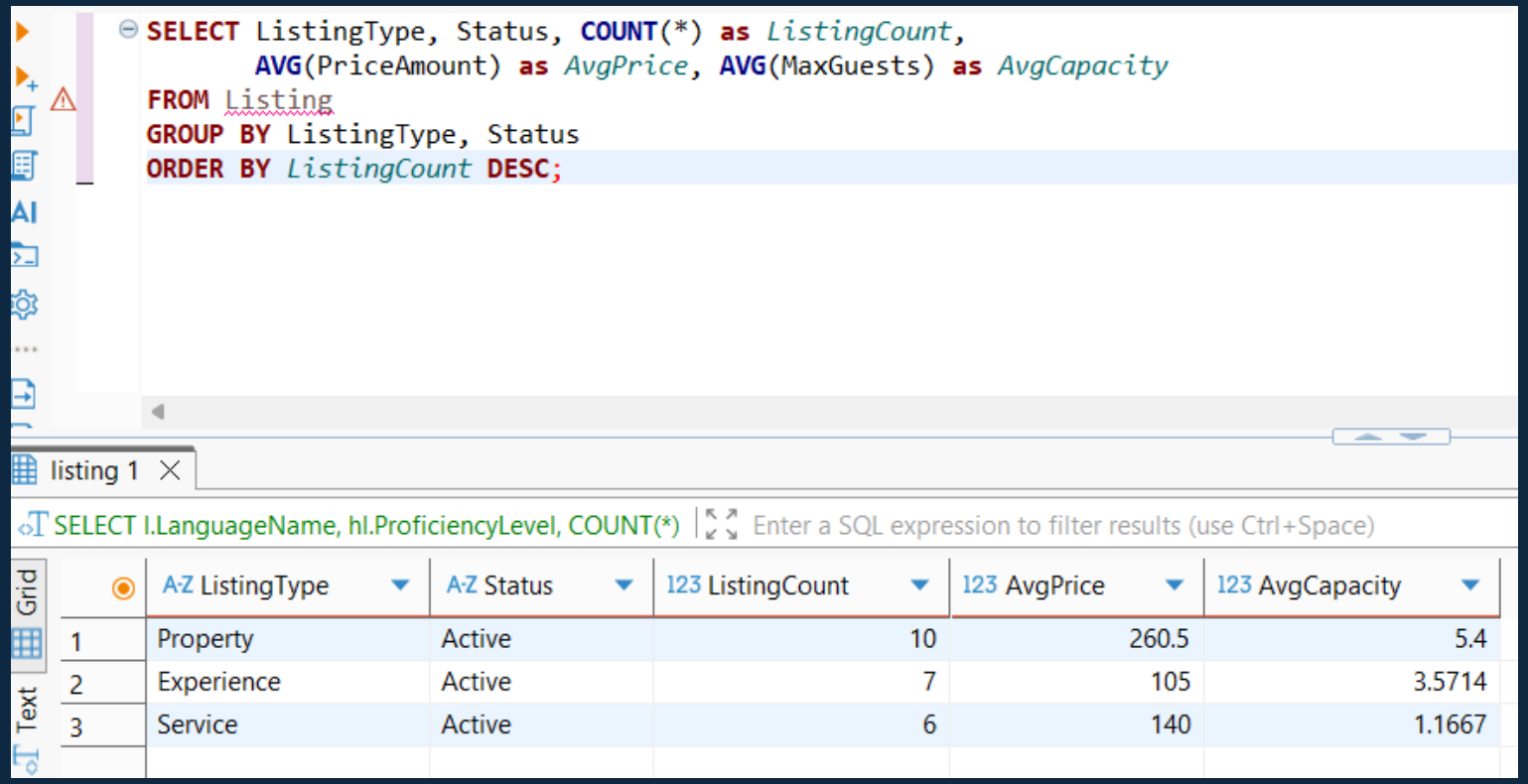
14 Listing

Purpose: Core listing information for properties, experiences, and services

Business Question: What is our listing portfolio composition and pricing strategy by type?

Query:

```
SELECT ListingType, Status, COUNT(*) as ListingCount,
      AVG(PriceAmount) as AvgPrice, AVG(MaxGuests) as AvgCapacity
FROM Listing
GROUP BY ListingType, Status
ORDER BY ListingCount DESC;
```



15 AvailabilityCalendar

Purpose: Daily availability tracking for listings to control bookable dates

Business Question: How much booking capacity do we have available in the next 30 days?

Query:

```
SELECT DATE(AvailableDate) as BookingDate,
    COUNT(*) as AvailableListings,
    SUM(CASE WHEN IsAvailable = 1 THEN 1
ELSE 0 END) as OpenSlots
FROM Availability_Calendar
WHERE AvailableDate BETWEEN CURDATE()
AND DATE_ADD(CURDATE(), INTERVAL 30
DAY)
GROUP BY DATE(AvailableDate)
ORDER BY BookingDate
LIMIT 10;
```

```
SELECT DATE(AvailableDate) as BookingDate,
       COUNT(*) as AvailableListings,
       SUM(CASE WHEN IsAvailable = 1 THEN 1 ELSE 0 END) as OpenSlots
FROM Availability_Calendar
GROUP BY DATE(AvailableDate)
ORDER BY BookingDate
LIMIT 10;
```

Results 1

SELECT DATE(AvailableDate) as BookingDate, COUNT(* | Enter a SQL expression to filter results (use

	BookingDate	AvailableListings	OpenSlots
1	2026-01-15	1	1
2	2026-01-16	1	1
3	2026-01-17	1	1
4	2026-01-18	1	1
5	2026-01-19	1	0
6	2026-01-20	1	1
7	2026-01-21	1	1
8	2026-01-22	4	4
9	2026-01-23	4	3
10	2026-01-24	4	2

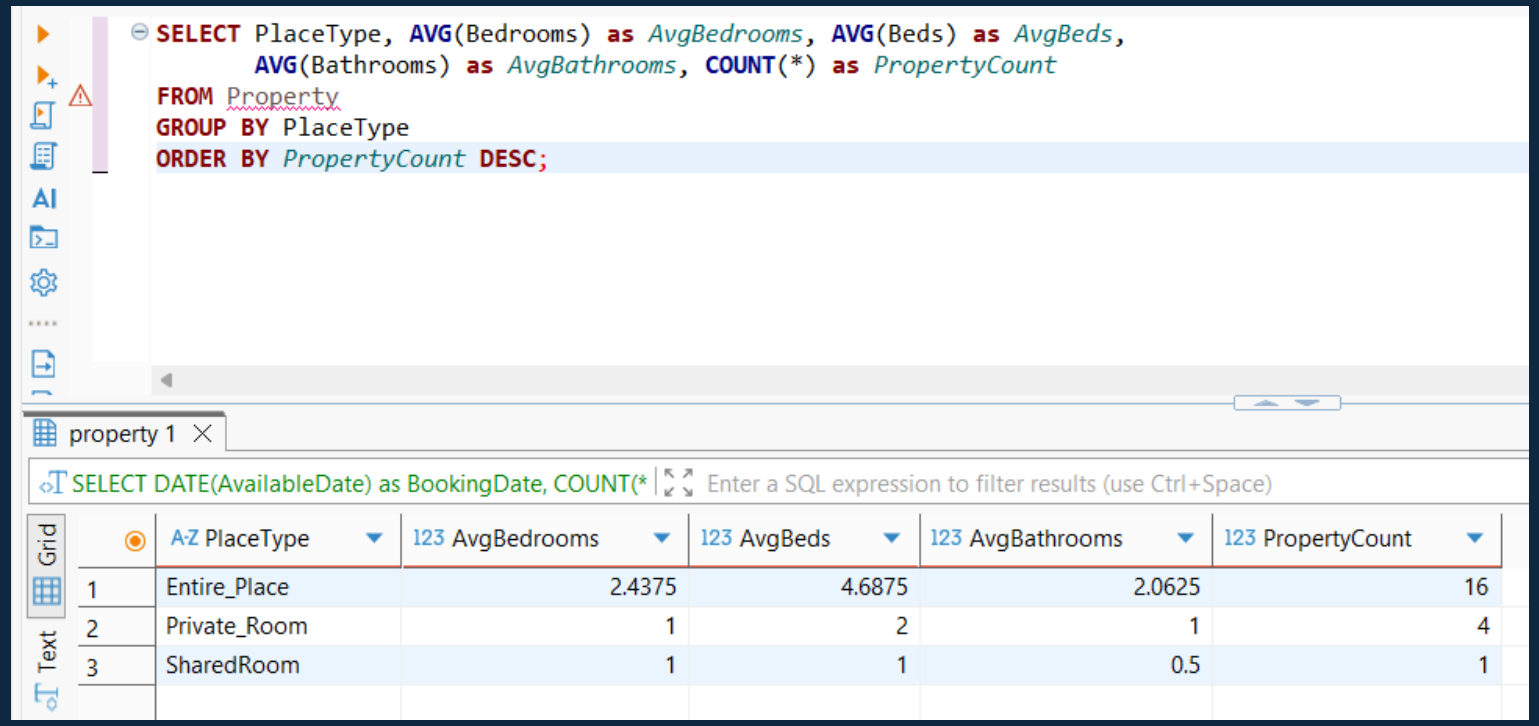
16 PropertyType

Purpose: Accommodation-specific details including rooms, bathrooms, and check-in policies

Business Question: What are the accommodation characteristics and capacity distribution of our properties?

Query:

```
SELECT PlaceType,
AVG(Bedrooms) as AvgBedrooms,
AVG(Beds) as AvgBeds,
    AVG(Bathrooms) as
AvgBathrooms, COUNT(*) as
PropertyCount
FROM Property
GROUP BY PlaceType
ORDER BY PropertyCount DESC;
```



	PlaceType	AvgBedrooms	AvgBeds	AvgBathrooms	PropertyCount
1	Entire_Place	2.4375	4.6875	2.0625	16
2	Private_Room	1	2	1	4
3	SharedRoom	1	1	0.5	1

17 ExperienceType

Purpose: Experience-specific details including duration, age requirements, and difficulty

Business Question: When are experiences typically offered and what difficulty levels do we provide?"

Query:

```
SELECT TimeOfDay, PhysicalDemand, COUNT(*) as ExperienceCount,
    AVG(DurationMinutes) as AvgDuration, AVG(MinAge) as AvgMinAge
FROM Experience
GROUP BY TimeOfDay, PhysicalDemand;
```

```
SELECT TimeOfDay, PhysicalDemand, COUNT(*) as ExperienceCount,
       AVG(DurationMinutes) as AvgDuration, AVG(MinAge) as AvgMinAge
FROM Experience
GROUP BY TimeOfDay, PhysicalDemand;
```

experience 1

SELECT PlaceType, AVG(Bedrooms) as AvgBedrooms, A\ | Enter a SQL expression to filter results (use Ctrl+Space)

	TimeOfDay	PhysicalDemand	ExperienceCount	AvgDuration	AvgMinAge
1	EVENING	LOW	5	144	13.8
2	MORNING	MODERATE	2	150	14
3	FLEXIBLE	LOW	3	110	16.6667
4	AFTERNOON	MODERATE	4	157.5	16.25
5	MORNING	HIGH	2	240	17
6	FLEXIBLE	MODERATE	2	210	18.5
7	MORNING	LOW	3	130	12
8	AFTERNOON	LOW	1	90	16
9	FLEXIBLE	HIGH	1	180	18
10	EVENING	MODERATE	1	120	16

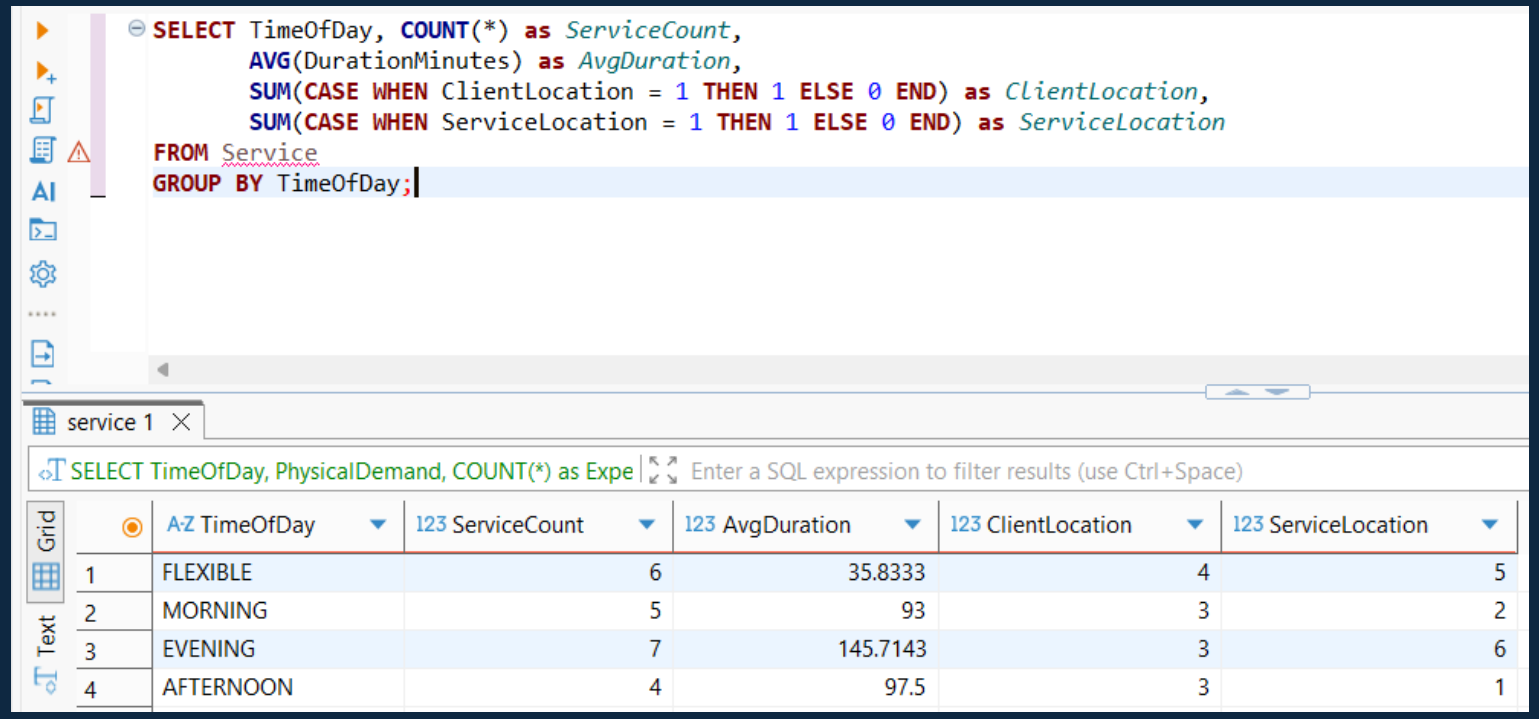
18 Service

Purpose: Service-specific details including location flexibility and equipment requirements

Business Question: What are the delivery patterns and location flexibility of our services?

Query:

```
SELECT TimeOfDay, COUNT(*) as ServiceCount,
    AVG(DurationMinutes) as AvgDuration,
    SUM(CASE WHEN ClientLocation = 1 THEN 1 ELSE 0 END) as ClientLocation,
    SUM(CASE WHEN ServiceLocation = 1 THEN 1 ELSE 0 END) as ServiceLocation
FROM Service
GROUP BY TimeOfDay;
```



	TimeOfDay	ServiceCount	AvgDuration	ClientLocation	ServiceLocation
1	FLEXIBLE	6	35.8333	4	5
2	MORNING	5	93	3	2
3	EVENING	7	145.7143	3	6
4	AFTERNOON	4	97.5	3	1

19 ListingPhotos

Purpose: Photo management for listings with categorization and display ordering

Business Question: How complete is our visual marketing across different photo categories?

Query:

```
SELECT PhotoCategory,
COUNT(*) as PhotoCount,
    SUM(CASE WHEN
IsMainPhoto = 1 THEN 1 ELSE 0
END) as MainPhotos
FROM ListingPhotos
GROUP BY PhotoCategory
ORDER BY PhotoCount DESC;
```

```
SELECT PhotoCategory, COUNT(*) as PhotoCount,
       SUM(CASE WHEN IsMainPhoto = 1 THEN 1 ELSE 0 END) as MainPhotos
FROM ListingPhotos
GROUP BY PhotoCategory
ORDER BY PhotoCount DESC;
```

listingphotos 1

SELECT TimeOfDay, COUNT(*) as ServiceCount, AVG(Du

Enter a SQL expression to filter results (use Ctrl+Space)

	PhotoCategory	PhotoCount	MainPhotos
1	main	17	17
2	bedroom	8	0
3	outdoor	7	0
4	bathroom	2	0
5	kitchen	1	0

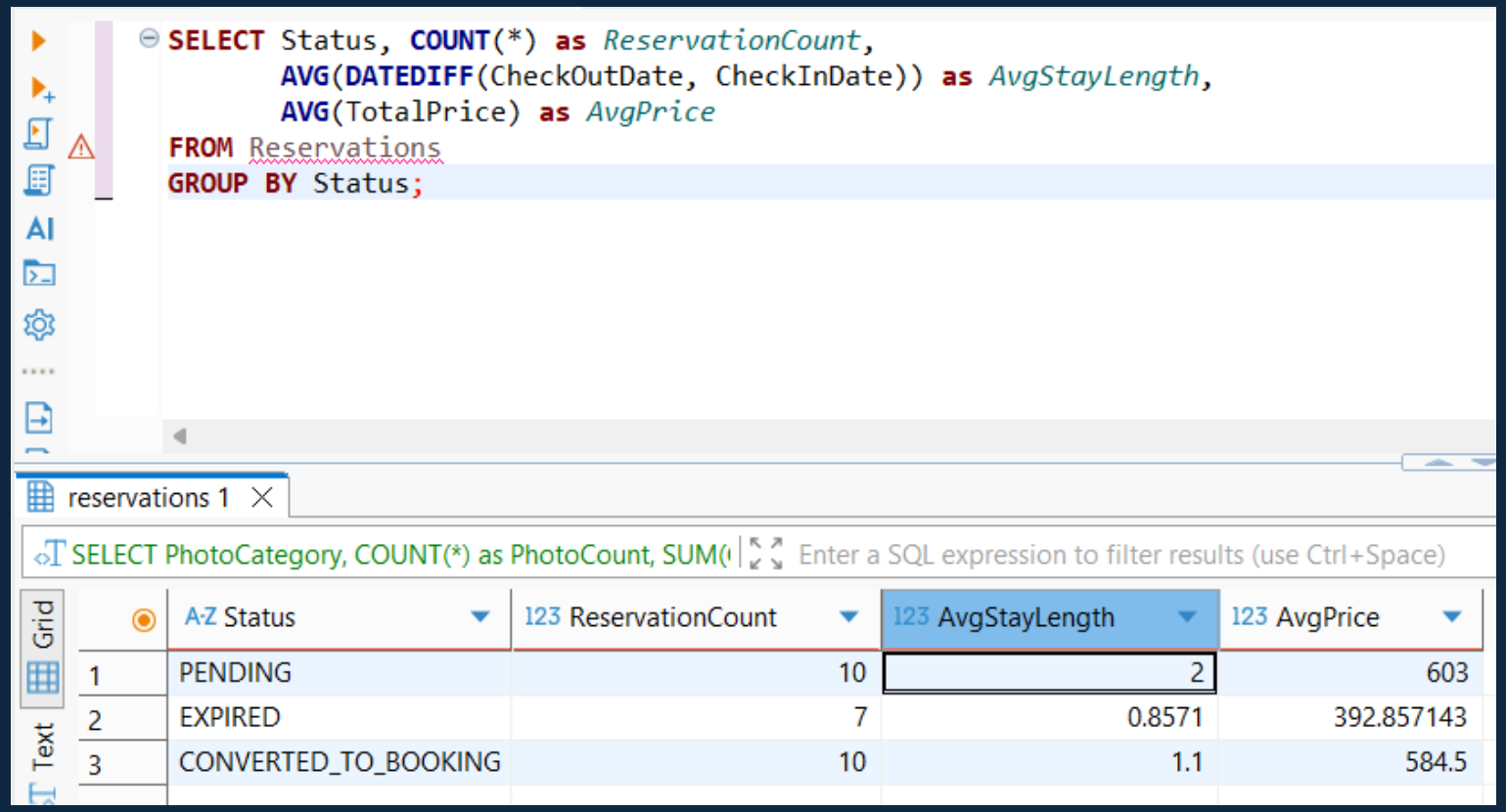
20 Reservations

Purpose: Temporary booking holds with 24-hour expiry before conversion to confirmed bookings

Business Question: How effective is our reservation system and what are typical booking patterns?

Query:

```
SELECT Status, COUNT(*) as ReservationCount,
    AVG(DATEDIFF(CheckOutDate, CheckInDate)) as AvgStayLength,
    AVG(TotalPrice) as AvgPrice
FROM Reservations
GROUP BY Status;
```



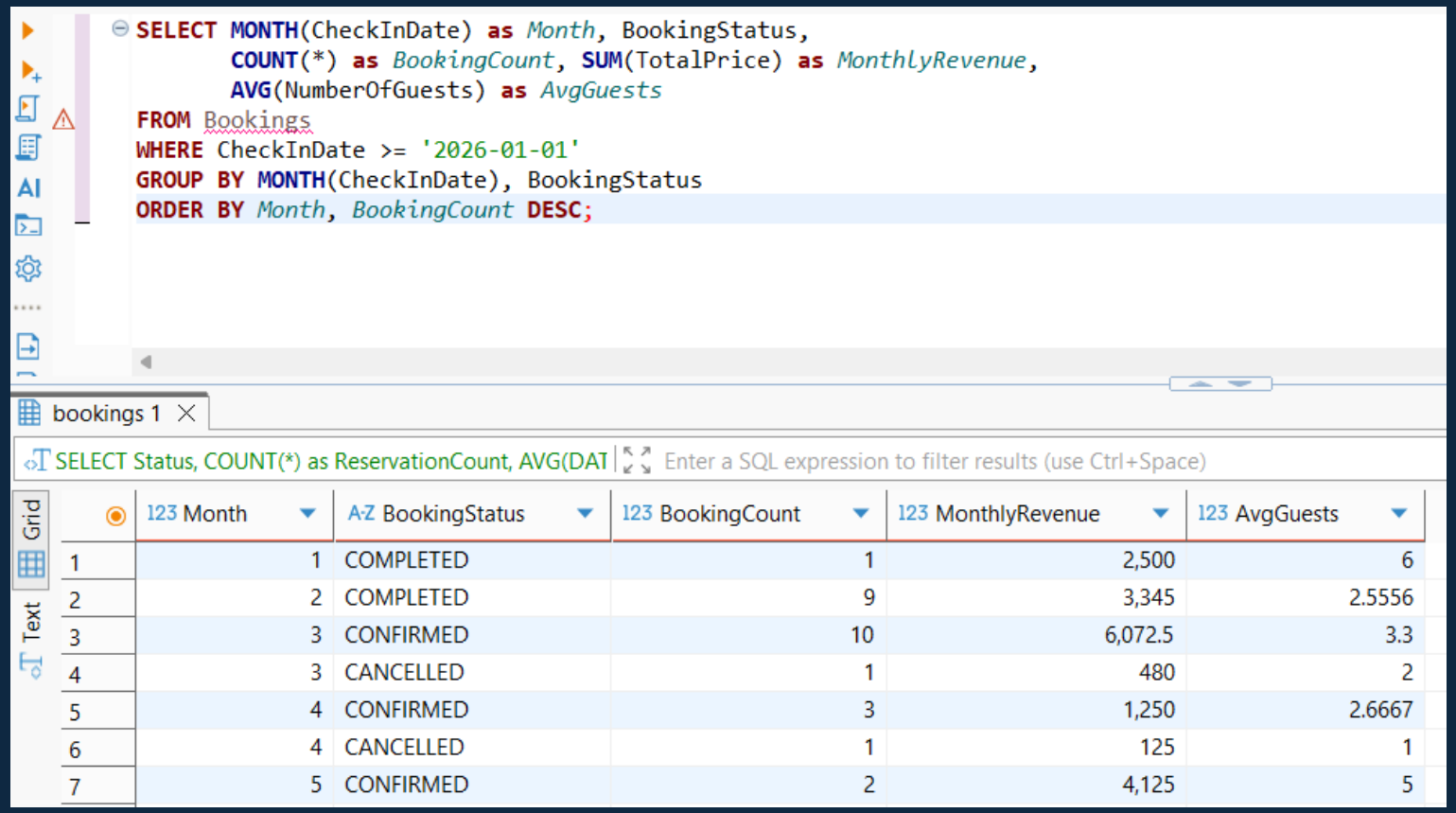
21 Bookings

Purpose: Confirmed bookings with payment - the core transaction entity

Business Question: What are our monthly booking trends and revenue patterns by status?

Query:

```
SELECT MONTH(CheckInDate) as Month, BookingStatus,
    COUNT(*) as BookingCount, SUM(TotalPrice) as MonthlyRevenue,
    AVG(NumberOfGuests) as AvgGuests
FROM Bookings
WHERE CheckInDate >= '2026-01-01'
GROUP BY MONTH(CheckInDate), BookingStatus
ORDER BY Month, BookingCount DESC;
```



	123 Month	A-Z BookingStatus	123 BookingCount	123 MonthlyRevenue	123 AvgGuests
1	1	COMPLETED	1	2,500	6
2	2	COMPLETED	9	3,345	2.5556
3	3	CONFIRMED	10	6,072.5	3.3
4	3	CANCELLED	1	480	2
5	4	CONFIRMED	3	1,250	2.6667
6	4	CANCELLED	1	125	1
7	5	CONFIRMED	2	4,125	5

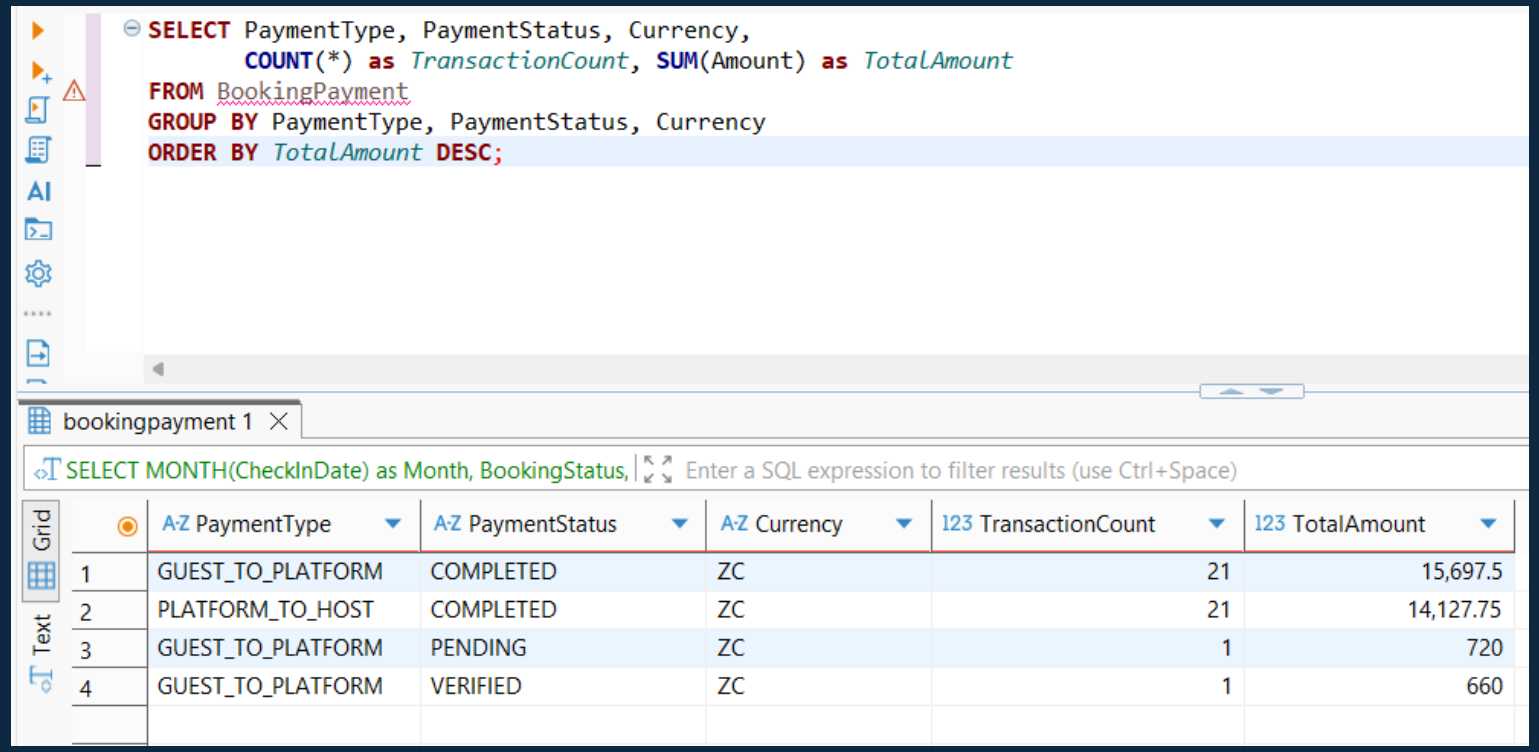
22 BookingPayment

Purpose: Financial transactions tracking payments from guests to platform to hosts

Business Question: How is our payment processing performing and where is revenue flowing?

Query:

```
SELECT PaymentType,
PaymentStatus, Currency,
    COUNT(*) as TransactionCount,
SUM(Amount) as TotalAmount
FROM BookingPayment
GROUP BY PaymentType,
PaymentStatus, Currency
ORDER BY TotalAmount DESC;
```



	PaymentType	PaymentStatus	Currency	TransactionCount	TotalAmount
1	GUEST_TO_PLATFORM	COMPLETED	ZC	21	15,697.5
2	PLATFORM_TO_HOST	COMPLETED	ZC	21	14,127.75
3	GUEST_TO_PLATFORM	PENDING	ZC	1	720
4	GUEST_TO_PLATFORM	VERIFIED	ZC	1	660

23 Cancellations

Purpose: Cancelled bookings with refund processing and reason tracking

Business Question: Who is cancelling bookings and what is the financial impact of refunds?

Query:

```
SELECT CancelledBy, RefundStatus, COUNT(*) as CancellationCount,
    AVG(RefundAmount) as AvgRefundAmount, SUM(RefundAmount) as TotalRefunds
FROM Cancellations
GROUP BY CancelledBy, RefundStatus
ORDER BY CancellationCount DESC;
```

```
SELECT CancelledBy, RefundStatus, COUNT(*) as CancellationCount,
       AVG(RefundAmount) as AvgRefundAmount, SUM(RefundAmount) as TotalRefunds
FROM Cancellations
GROUP BY CancelledBy, RefundStatus
ORDER BY CancellationCount DESC;
```

cancellations 1

SELECT PaymentType, PaymentStatus, Currency, COUNT(Enter a SQL expression to filter results (use Ctrl+Space)

	CancelledBy	RefundStatus	CancellationCount	AvgRefundAmount	TotalRefunds
1	GUEST	PROCESSED	7	235.714286	1,650
2	HOST	PROCESSED	7	594.642857	4,162.5
3	ADMIN	PROCESSED	6	859.166667	5,155
4	GUEST	PENDING	3	782.5	2,347.5
5	GUEST	REJECTED	2	0	0
6	ADMIN	REJECTED	2	0	0

24 Reviews

Purpose: Guest feedback system for rating hosts and overall experience quality

Business Question: What is our overall guest satisfaction and how are hosts performing?

Query:

```
SELECT OverallRating, COUNT(*) as ReviewCount,
    AVG(HostRating) as AvgHostRating
FROM Reviews
GROUP BY OverallRating
ORDER BY OverallRating DESC;
```

```
SELECT OverallRating, COUNT(*) as ReviewCount,
       AVG(HostRating) as AvgHostRating
FROM Reviews
GROUP BY OverallRating
ORDER BY OverallRating DESC;
```

reviews 1

SELECT DATE(AvailableDate) as BookingDate, COUNT(*

Enter a SQL expression to filter re

	123 OverallRating	123 ReviewCount	123 AvgHostRating
1	5	8	4.76875
2	4	11	4.15
3	3	6	3.083333

25 Messages(Recursive Relationship)

Purpose: Communication system with threaded conversations between guests and hosts

Business Question: Which bookings generate the most communication and how are conversations structured?

Query:

```
SELECT b.BookingID,
COUNT(m.MessageID) as MessageCount,
    COUNT(CASE WHEN
m.ReplyToMessageID IS NULL THEN 1
END) as NewThreads,
    COUNT(CASE WHEN
m.ReplyToMessageID IS NOT NULL THEN
1 END) as Replies,
    MAX(m.SentAt) as LastMessage
FROM Messages m
JOIN Bookings b ON m.BookingID =
b.BookingID
GROUP BY b.BookingID
ORDER BY MessageCount DESC
LIMIT 10;
```

```
SELECT b.BookingID, COUNT(m.MessageID) as MessageCount,
       COUNT(CASE WHEN m.ReplyToMessageID IS NULL THEN 1 END) as NewThreads,
       COUNT(CASE WHEN m.ReplyToMessageID IS NOT NULL THEN 1 END) as Replies,
       MAX(m.SentAt) as LastMessage
FROM Messages m
JOIN Bookings b ON m.BookingID = b.BookingID
GROUP BY b.BookingID
ORDER BY MessageCount DESC
LIMIT 10;
```

bookings 1

SELECT OverallRating, COUNT(*) as ReviewCount, AVG(H

Enter a SQL expression to filter results (use Ctrl+Space)

	BookingID	MessageCount	NewThreads	Replies	LastMessage
1	147	4	2	2	2026-02-17 13:45:00.0
2	150	4	2	2	2026-02-24 18:30:00.0
3	146	2	1	1	2026-02-14 11:45:00.0
4	149	2	1	1	2026-02-21 12:30:00.0
5	148	2	1	1	2026-02-19 18:45:00.0
6	152	2	1	1	2026-02-28 16:45:00.0
7	153	2	1	1	2026-03-02 11:30:00.0
8	154	2	1	1	2026-03-05 12:15:00.0
9	155	2	1	1	2026-03-08 13:20:00.0
10	151	2	1	1	2026-02-26 11:45:00.0

26 Payment Account

Purpose: Ternary relationship linking hosts, listings, and banking details for payments

Business Question: How do hosts manage their banking setup and what is their verification success rate?

Query:

```
SELECT AccountType,
VerificationStatus, COUNT(*) as
AccountCount,
    SUM(CASE WHEN IsPrimary = 1
THEN 1 ELSE 0 END) as
PrimaryAccounts,
    SUM(CASE WHEN IsActive = 1
THEN 1 ELSE 0 END) as
ActiveAccounts
FROM Payment_Account
GROUP BY AccountType,
VerificationStatus
ORDER BY AccountCount DESC;
```

```
SELECT AccountType, VerificationStatus, COUNT(*) as AccountCount,
       SUM(CASE WHEN IsPrimary = 1 THEN 1 ELSE 0 END) as PrimaryAccounts,
       SUM(CASE WHEN IsActive = 1 THEN 1 ELSE 0 END) as ActiveAccounts
FROM Payment_Account
GROUP BY AccountType, VerificationStatus
ORDER BY AccountCount DESC;
```

payment_account 1

SELECT b.BookingID, COUNT(m.MessageID) as Message | Enter a SQL expression to filter results (use Ctrl+Space)

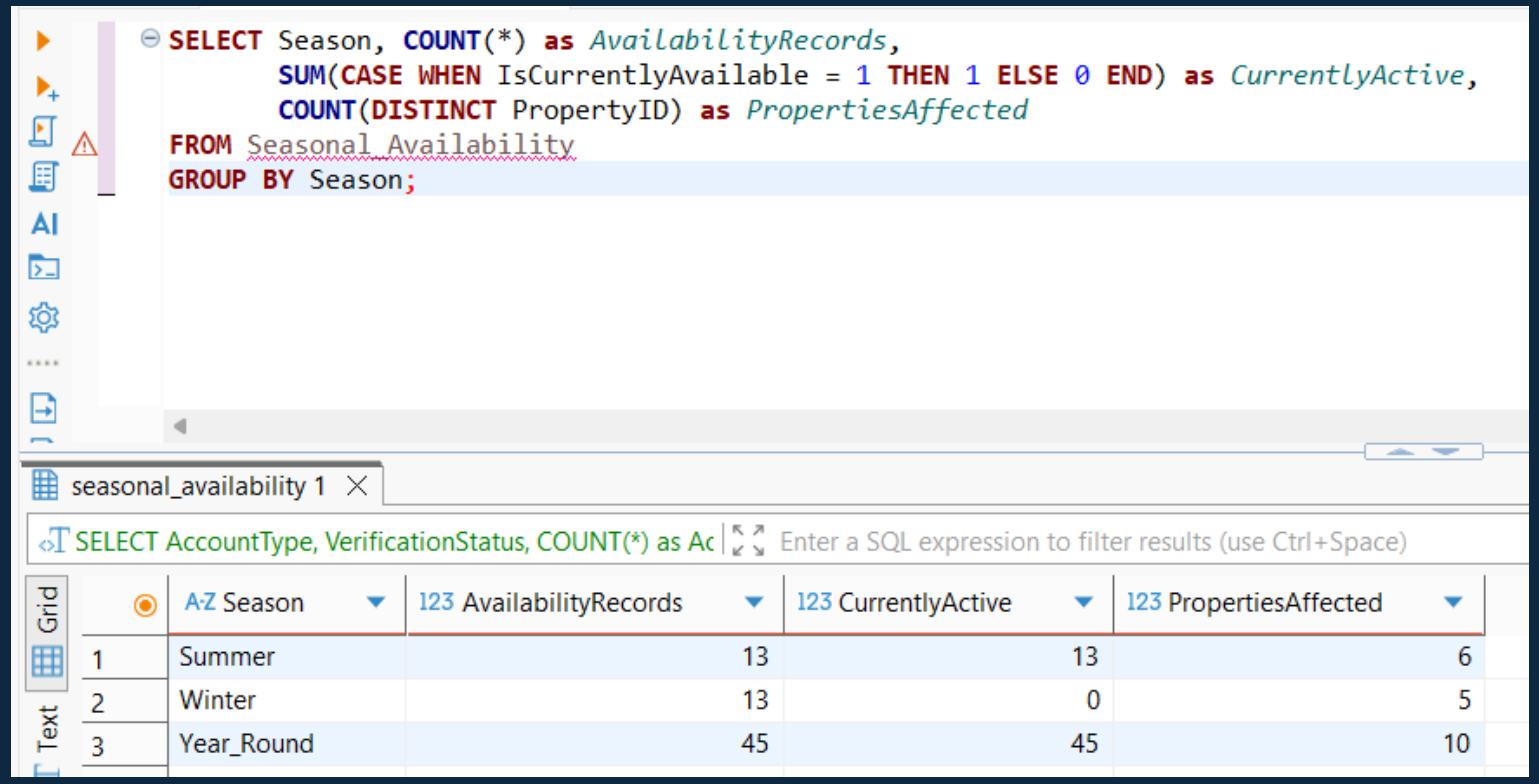
	AccountType	VerificationStatus	AccountCount	PrimaryAccounts	ActiveAccounts
1	Business	VERIFIED	16	5	16
2	Personal	VERIFIED	13	4	13
3	Personal	PENDING	3	0	2
4	Business	PENDING	1	0	1
5	Personal	FAILED	1	0	0
6	Personal	REJECTED	1	0	0

26 Seasonal_Availability

Ternary Relationship

Purpose: Ternary relationship tracking seasonal availability of amenities across properties

Business Question: How do seasonal changes affect amenity availability across our properties?

Query:

```
SELECT Season, COUNT(*) as AvailabilityRecords,
    SUM(CASE WHEN IsCurrentlyAvailable = 1 THEN 1 ELSE 0 END) as CurrentlyActive,
    COUNT(DISTINCT PropertyID) as PropertiesAffected
FROM Seasonal_Availability
GROUP BY Season;
```